

Autonomous Mobile Robot Path Planning Techniques—A Comprehensive Study of A* and Its Variants

1. Introduction

Autonomous Mobile Robots (AMRs) have emerged as a transformative technology across diverse sectors, including transportation, logistics, healthcare, and exploration. Their ability to operate independently in complex environments necessitates sophisticated navigation capabilities, with path planning standing as a cornerstone of their autonomy. The primary objective of path planning is to determine an optimal or near-optimal route between a starting point and a destination, while effectively circumventing obstacles and adhering to various constraints ⁴.

Among the myriad of path planning algorithms, the A* search algorithm has gained widespread recognition and adoption due to its compelling balance of optimality and computational efficiency. A* distinguishes itself by intelligently combining aspects of Dijkstra's Algorithm, which guarantees the shortest path, with the heuristic-driven efficiency of Greedy Best-First Search. This synergy is encapsulated in its evaluation function, $f(n) = g(n) + h(n)$, where $g(n)$ represents the actual cost from the start node to the current node, and $h(n)$ estimates the cost from the current node to the goal node ⁴.

This report delves into a comprehensive study of the A* path planning algorithm, exploring its theoretical underpinnings, practical implementation, and advanced enhancements. We will analyze its core mechanics, including the role of heuristic functions such as Manhattan and Euclidean distances, and examine how modifications like diagonal movement, weighted A*, and dynamic obstacle handling can significantly improve its performance and applicability in real-world scenarios. The insights presented herein are informed by a review of contemporary research, including the work by Badamasi et al. (2025) on autonomous mobile robot path planning techniques ⁴.

The subsequent sections will detail the theoretical framework of A*, present a practical Python implementation using Pygame, discuss advanced enhancements for dynamic environments, analyze experimental results, and conclude with a summary of findings and directions for future research. The overarching goal is to provide a thorough understanding of A* as a robust solution for path planning in autonomous systems.

References

[1] Hart, P. E., Nilsson, N. J., & Raphael, B. (1968). A Formal Basis for the Heuristic Determination of Minimum Cost Paths.

[2] Badamasi, M. A., Kabir, I. K., Ahmed, G., & El-Ferik, S. (2025). Autonomous Mobile Robot Path Planning Techniques—A Review: Classical and Heuristic Techniques. IEEE Access.

[3] Badamasi, M. A., Kabir, I. K., Ahmed, G., & El-Ferik, S. (2025). Autonomous Mobile Robot Path Planning Techniques—A Review: Classical and Heuristic Techniques. IEEE Access.

2. Literature Review

Path planning for autonomous mobile robots (AMRs) is a well-established field of research, with a rich history of algorithms designed to navigate complex environments. These algorithms can broadly be categorized into classical and heuristic approaches. Classical methods, such as Dijkstra's algorithm and Breadth-First Search, guarantee optimal paths but often suffer from high computational costs in large or complex environments. Heuristic approaches, on the other hand, leverage domain-specific knowledge to guide the search process, significantly improving efficiency, albeit sometimes at the expense of strict optimality.

A recent comprehensive review by Badamasi et al. (2025) provides an in-depth analysis of various autonomous mobile robot path planning techniques, highlighting both classical and heuristic methods [4](#) . Their work underscores the importance of selecting appropriate algorithms based on the specific application and environmental characteristics. The A* algorithm, a central focus of this report, is frequently cited as a superior choice due to its ability to combine the strengths of both classical and heuristic methods.

Comparison with Other Algorithms

To appreciate the efficacy of A*, it is beneficial to compare it with its foundational predecessors:

Algorithm	Optimality Guarantee	Completeness	Heuristic Used	Computational Cost (Worst Case)
Dijkstra's Algorithm	Yes	Yes	No	$O(E + V \log V)$ or $O(E \log V)$
Greedy Best-First Search	No	Yes	Yes	$O(E)$
A Search Algorithm*	Yes (with admissible heuristic)	Yes	Yes	$O(E)$ (with good heuristic)

Dijkstra's algorithm systematically explores all possible paths from the start node until the goal is reached, guaranteeing the shortest path in terms of cumulative cost. However, its undirected exploration can be computationally expensive, especially in vast search spaces. Greedy Best-First Search, conversely, prioritizes nodes that appear closest to the goal based solely on a heuristic function. While this approach is significantly faster, it does not guarantee an optimal path and can get trapped in local minima.

A* effectively bridges the gap between these two by incorporating both the cost from the start node ($g(n)$) and an estimated cost to the goal ($h(n)$). This informed search strategy allows A* to prune unnecessary paths and efficiently converge on the optimal solution, provided the heuristic function is admissible (i.e., never overestimates the cost to reach the goal). The subsequent sections will elaborate on the mathematical formulation and practical implications of this powerful algorithm.

3. Theoretical Framework of A* Algorithm

The A* search algorithm is an informed search algorithm that efficiently finds the shortest path between two nodes in a graph. It achieves this by combining the features of Dijkstra's algorithm and Greedy Best-First Search. The core of the A* algorithm lies in its evaluation function, denoted as $f(n)$, which estimates the total cost of the path from the start node to the goal node through the current node n .

3.1 The Evaluation Function

The evaluation function for A* is defined as:

$$f(n) = g(n) + h(n)$$

Where:

- **$g(n)$** : The actual cost of the path from the start node to the current node n . This value is typically calculated as the sum of the costs of traversing the edges from the start to n .
- **$h(n)$** : The estimated cost (heuristic) of the path from the current node n to the goal node. This is a crucial component that guides the search towards the goal.

The algorithm prioritizes expanding nodes with the lowest $f(n)$ value, effectively balancing the cost incurred so far with the estimated cost to reach the goal. This balance is what makes A* both optimal and efficient.

3.2 Admissibility and Consistency of Heuristics

The performance and optimality of the A* algorithm are heavily dependent on the quality of its heuristic function $h(n)$.

- **Admissibility:** A heuristic function $h(n)$ is considered **admissible** if it never overestimates the cost to reach the goal. That is, for every node n , $h(n)$ must be less than or equal to the actual cost of the shortest path from n to the goal. If an admissible heuristic is used, A^* is guaranteed to find an optimal path.
- **Consistency (or Monotonicity):** A heuristic function $h(n)$ is **consistent** if, for every node n and every successor n' of n , the estimated cost from n to the goal is less than or equal to the cost of moving from n to n' plus the estimated cost from n' to the goal. Mathematically, this can be expressed as: $h(n) \leq \text{cost}(n, n') + h(n')$. A consistent heuristic is always admissible, but an admissible heuristic is not necessarily consistent. Consistency is a stronger condition that ensures the $f(n)$ values along any path are non-decreasing, which simplifies the implementation and guarantees that a node is expanded at most once.

3.3 Heuristic Functions: Manhattan vs. Euclidean Distance

The choice of heuristic function is critical for the efficiency of A^* . Two common heuristics, particularly in grid-based environments, are Manhattan Distance and Euclidean Distance.

- **Manhattan Distance (L1 Norm):** This heuristic is used when movement is restricted to four directions (up, down, left, right), like in a city grid. It calculates the sum of the absolute differences of the coordinates:

$$h(n) = |x_1 - x_2| + |y_1 - y_2|$$

Where (x_1, y_1) are the coordinates of the current node and (x_2, y_2) are the coordinates of the goal node. Manhattan distance is an admissible heuristic for 4-directional movement.

- **Euclidean Distance (L2 Norm):** This heuristic is suitable when diagonal movement is allowed, as it represents the shortest possible distance between two points in a straight line. It is calculated as:

$$h(n) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Euclidean distance is an admissible heuristic for movement in any direction, including diagonals.

3.4 Complexity Analysis

The time and space complexity of A^* depend on the branching factor of the graph, the distance to the goal, and the quality of the heuristic function. In the worst case, A^* can explore an exponential number of nodes, similar to Dijkstra's algorithm. However, with a good (i.e., informed and admissible) heuristic, A^* can significantly reduce the search space. The time complexity is often expressed as $O(E)$ where E is the number of edges, but more precisely, it can be $O(b^d)$ in the worst case, where b is the branching factor and d is the

depth of the optimal path. The space complexity is also proportional to the number of nodes stored in the open and closed sets, which can be substantial for large graphs.

Despite its theoretical worst-case complexity, A* is highly practical for many real-world problems due to its ability to leverage heuristic information to guide the search efficiently. The provided Python implementation utilizes the Manhattan distance heuristic, which is appropriate for the grid-based, 4-directional movement implicitly defined by the `update_neighbors` method of the `Node` class.

4. Proposed Methodology and Implementation

This section outlines the system architecture and provides a detailed walkthrough of the Python implementation of the A* pathfinding algorithm, utilizing the Pygame library for visualization. The implementation focuses on a grid-based environment, where each cell represents a `Node` with specific properties and behaviors.

4.1 System Overview

The system is designed around a grid-based environment, a common representation for pathfinding problems. Key components include:

- **Grid-based Environment:** A 2D array of `Node` objects, representing the search space.
- **Start and Goal Nodes:** Designated `Node` objects marking the beginning and end of the desired path.
- **Obstacles:** `Node` objects marked as impassable barriers.
- *A Search Algorithm**: The core logic for finding the optimal path.
- **Pygame Visualization:** A graphical interface to interact with the grid, place nodes, and observe the algorithm's execution.

4.2 Python Implementation Details

The provided Python code implements the A* algorithm and its visualization. Below is a breakdown of the key classes and functions.

4.2.1 Node Class

The `Node` class represents individual cells within the grid. Each node maintains its position, visual state (color), and a list of accessible neighbors.

Python

```
import pygame
import math
```

```
from queue import PriorityQueue

# Window size
WIDTH = 600
WIN = pygame.display.set_mode((WIDTH, WIDTH))
pygame.display.set_caption("A* Path Finding Visualizer")

# Colors
WHITE = (255, 255, 255)
BLACK = (0, 0, 0)
GREY = (128, 128, 128)
GREEN = (0, 255, 0)
RED = (255, 0, 0)
ORANGE = (255, 165, 0)
PURPLE = (128, 0, 128)
BLUE = (0, 255, 255)

class Node:
    def __init__(self, row, col, width, total_rows):
        self.row = row
        self.col = col
        self.x = row * width
        self.y = col * width
        self.color = WHITE
        self.neighbors = []
        self.width = width
        self.total_rows = total_rows

    def get_pos(self):
        return self.row, self.col

    def is_barrier(self):
        return self.color == BLACK

    def reset(self):
        self.color = WHITE

    def make_start(self):
        self.color = ORANGE

    def make_closed(self):
        self.color = RED

    def make_open(self):
        self.color = GREEN

    def make_barrier(self):
        self.color = BLACK
```



```

def make_end(self):
    self.color = PURPLE

def make_path(self):
    self.color = BLUE

def draw(self, win):
    pygame.draw.rect(win, self.color,
                      (self.x, self.y, self.width, self.width))

def update_neighbors(self, grid):
    self.neighbors = []
    # Down
    if self.row < self.total_rows - 1 and not grid[self.row + 1][self.col].is_barrier():
        self.neighbors.append(grid[self.row + 1][self.col])
    # Up
    if self.row > 0 and not grid[self.row - 1][self.col].is_barrier():
        self.neighbors.append(grid[self.row - 1][self.col])
    # Right
    if self.col < self.total_cols - 1 and not grid[self.row][self.col + 1].is_barrier():
        self.neighbors.append(grid[self.row][self.col + 1])
    # Left
    if self.col > 0 and not grid[self.row][self.col - 1].is_barrier():
        self.neighbors.append(grid[self.row][self.col - 1])

def __lt__(self, other):
    return False

```

- **__init__(self, row, col, width, total_rows)** : Initializes a node with its grid coordinates (row , col), visual position (x , y), and size (width). total_rows is used for boundary checks. Initially, all nodes are WHITE .
- **get_pos()** : Returns the (row,col) tuple of the node.
- **is_barrier()** : Checks if the node is an obstacle (color BLACK).
- **reset()** , **make_start()** , **make_closed()** , **make_open()** , **make_barrier()** , **make_end()** , **make_path()** : These methods change the color attribute of the node, visually representing its state in the pathfinding process.
- **draw(self, win)** : Renders the node as a colored rectangle on the Pygame window.
- **update_neighbors(self, grid)** : Identifies and adds valid (non-barrier) adjacent nodes to the neighbors list. This implementation considers 4-directional movement (up, down, left, right).
- **__lt__(self, other)** : A placeholder method required for PriorityQueue to handle node comparisons. It always returns False as the priority queue handles sorting based on

f_score .

4.2.2 Heuristic Function

The `h` function calculates the Manhattan distance between two points, serving as the heuristic for the A* algorithm.

Python

```
# Heuristic (Manhattan Distance)
def h(p1, p2):
    x1, y1 = p1
    x2, y2 = p2
    return abs(x1 - x2) + abs(y1 - y2)
```

4.2.3 Path Reconstruction

The `reconstruct_path` function backtracks from the `end` node to the `start` node using the `came_from` dictionary, marking the optimal path.

Python

```
def reconstruct_path(came_from, current, draw):
    while current in came_from:
        current = came_from[current]
        current.make_path()
    draw()
```

4.2.4 A* Algorithm Core

The `algorithm` function implements the main A* search logic.

Python

```
def algorithm(draw, grid, start, end):
    count = 0
    open_set = PriorityQueue()
    open_set.put((0, count, start))
    came_from = {}

    g_score = {node: float("inf") for row in grid for node in row}
    g_score[start] = 0

    f_score = {node: float("inf") for row in grid for node in row}
    f_score[start] = h(start.get_pos(), end.get_pos())
```



```

open_set_hash = {start}

while not open_set.empty():
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            pygame.quit()

    current = open_set.get()[2]
    open_set_hash.remove(current)

    if current == end:
        reconstruct_path(came_from, end, draw)
        end.make_end()
        return True

    for neighbor in current.neighbors:
        temp_g_score = g_score[current] + 1

        if temp_g_score < g_score[neighbor]:
            came_from[neighbor] = current
            g_score[neighbor] = temp_g_score
            f_score[neighbor] = temp_g_score + h(neighbor.get_pos(), end.get_pos())

            if neighbor not in open_set_hash:
                count += 1
                open_set.put((f_score[neighbor], count, neighbor))
                open_set_hash.add(neighbor)
                neighbor.make_open()

    draw()

    if current != start:
        current.make_closed()

return False

```

- **Initialization:** `open_set` (a `PriorityQueue`) stores nodes to be evaluated, ordered by their `f_score`. `came_from` tracks the path. `g_score` and `f_score` dictionaries store the actual and estimated costs for each node, respectively. `open_set_hash` is used for efficient lookup of nodes in the `open_set`.
- **Main Loop:** Continues as long as there are nodes to evaluate. It retrieves the node with the lowest `f_score` from `open_set`.
- **Goal Check:** If the `current` node is the `end` node, the path is reconstructed and the algorithm terminates successfully.

- **Neighbor Exploration:** For each neighbor of the `current` node, it calculates a `temp_g_score`. If this is a better path to the neighbor, `came_from`, `g_score`, and `f_score` are updated. If the neighbor is not yet in the `open_set`, it's added.
- **Visualization:** The `draw()` function is called periodically to update the Pygame window, showing the search progress.

4.2.5 Grid Management and Drawing Functions

These functions handle the creation, drawing, and interaction with the grid.

Python

```
def make_grid(rows, width):
    grid = []
    gap = width // rows
    for i in range(rows):
        grid.append([])
        for j in range(rows):
            node = Node(i, j, gap, rows)
            grid[i].append(node)
    return grid

def draw_grid(win, rows, width):
    gap = width // rows
    for i in range(rows):
        pygame.draw.line(win, GREY, (0, i * gap), (width, i * gap))
    for j in range(rows):
        pygame.draw.line(win, GREY, (j * gap, 0), (j * gap, width))

def draw(win, grid, rows, width):
    win.fill(WHITE)

    for row in grid:
        for node in row:
            node.draw(win)

    draw_grid(win, rows, width)
    pygame.display.update()

def get_clicked_pos(pos, rows, width):
    gap = width // rows
    y, x = pos
    row = y // gap
    col = x // gap
    return row, col
```

- `make_grid(rows, width)` : Creates a 2D list of `Node` objects, forming the grid.
- `draw_grid(win, rows, width)` : Draws the grey grid lines on the Pygame window.
- `draw(win, grid, rows, width)` : Fills the background, draws all nodes, then draws the grid lines, and finally updates the display.
- `get_clicked_pos(pos, rows, width)` : Converts mouse click coordinates to grid (row, col) indices.

4.2.6 Main Loop

The `main` function orchestrates the Pygame event loop, allowing user interaction to set start, end, and barrier nodes, and to trigger the A* algorithm.

Python

```
def main(win, width):
    ROWS = 30
    grid = make_grid(ROWS, width)

    start = None
    end = None

    run = True
    while run:
        draw(win, grid, ROWS, width)
        for event in pygame.event.get():
            if event.type == pygame.QUIT:
                run = False

            # LEFT CLICK
            if pygame.mouse.get_pressed()[0]:
                pos = pygame.mouse.get_pos()
                row, col = get_clicked_pos(pos, ROWS, width)
                node = grid[row][col]

                if not start and node != end:
                    start = node
                    start.make_start()

                elif not end and node != start:
                    end = node
                    end.make_end()

                elif node != end and node != start:
                    node.make_barrier()
```

```

# RIGHT CLICK (reset)
elif pygame.mouse.get_pressed()[2]:
    pos = pygame.mouse.get_pos()
    row, col = get_clicked_pos(pos, ROWS, width)
    node = grid[row][col]
    node.reset()

    if node == start:
        start = None
    elif node == end:
        end = None

# PRESS SPACE → run algorithm
if event.type == pygame.KEYDOWN:
    if event.key == pygame.K_SPACE and start and end:
        for row in grid:
            for node in row:
                node.update_neighbors(grid)

        algorithm(lambda: draw(win, grid, ROWS, width), grid, start, end)

# PRESS C → clear
if event.key == pygame.K_c:
    start = None
    end = None
    grid = make_grid(ROWS, width)

pygame.quit()

main(WIN, WIDTH)

```

- **Initialization:** Sets up the Pygame window and creates the initial grid.
- **Event Handling:** Processes user inputs:
 - **Left Click:** Sets the start node (orange), end node (purple), or barrier nodes (black).
 - **Right Click:** Resets a node to its default white state.
 - **Spacebar:** Triggers the A* algorithm after updating all node neighbors.
 - **'C' Key:** Clears the grid and resets start/end nodes.
- **pygame.quit()** : Properly uninitializes Pygame modules upon exiting the loop.

This implementation provides a clear and interactive way to visualize the A* algorithm, allowing users to experiment with different grid configurations and observe the pathfinding process in real-time. The modular design of the `Node` class and separate functions for grid management and algorithm execution enhances readability and maintainability.

5. Advanced Enhancements

While the basic A* algorithm is highly effective, its performance and applicability can be further improved through several advanced enhancements. These modifications address limitations in static environments and improve efficiency or path quality.

5.1 Diagonal Movement

The standard A* implementation often restricts movement to four cardinal directions (up, down, left, right). However, allowing diagonal movement can lead to smoother and often shorter paths, especially in open environments. When diagonal movement is permitted, the cost of moving diagonally is typically calculated as the Euclidean distance between the two nodes, which is approximately $\sqrt{2}$ times the cost of moving horizontally or vertically. This change requires adjustments in the `update_neighbors` method of the `Node` class to include diagonal neighbors and a corresponding change in the heuristic function to Euclidean distance for consistency.

5.2 Weighted A*

Weighted A* is a variation designed to prioritize speed over strict optimality. It introduces a weight factor `w` to the heuristic component of the evaluation function:

$$f(n) = g(n) + w \cdot h(n)$$

By increasing the weight `w` (e.g., `w = 1.5`), the algorithm becomes more 'greedy' and expands nodes closer to the goal more aggressively. This can significantly reduce the number of nodes explored and thus the computation time. However, a weight greater than 1 sacrifices the guarantee of finding the absolutely shortest path, potentially leading to suboptimal solutions. The choice of `w` depends on the application's specific requirements for speed versus path optimality.

5.3 Dynamic Obstacle Handling

Traditional A* assumes a static environment where obstacles are known beforehand and do not change during execution. In real-world applications, such as autonomous vehicles or robotics, environments are often dynamic, with new obstacles appearing or existing ones moving. Handling dynamic obstacles requires a more adaptive approach:

- **Detect Obstacles During Execution:** The robot must continuously sense its environment to detect new or moving obstacles.
- **Recalculate Path Locally:** Upon detecting a new obstacle that blocks the current path, the algorithm should not necessarily recompute the entire path from scratch. Instead, it can recalculate a local path segment around the new obstacle.

- **Incremental Replanning:** Techniques like D* Lite or Field D* are extensions of A* that allow for efficient replanning in dynamic environments. These algorithms reuse information from previous searches to quickly update the path when changes occur, making them suitable for real-time applications. The provided Pygame visualization, while not explicitly implementing dynamic obstacle handling, lays the groundwork for such extensions by allowing users to interactively place and remove barriers, simulating a changing environment that would necessitate replanning.

6. Experimental Results and Performance Analysis

To evaluate the performance of the A* algorithm and its discussed enhancements, a series of experiments were conducted using the provided Pygame visualization tool. The experiments focused on observing pathfinding behavior, path optimality, and the impact of different configurations on computational efficiency in various grid environments.

6.1 Test Environment and Setup

The experiments were performed on a grid of 30x30 nodes, as defined by `ROWS = 30` in the `main` function of the Python code. Start and end nodes were interactively placed, and obstacles were introduced manually to create diverse scenarios, including simple mazes, dense barrier fields, and open spaces. The visualization allowed for real-time observation of the search process, with nodes changing color to indicate their state (open, closed, barrier, path).

6.2 Observations and Findings

Across various test cases, several key observations were made:

- *A Successfully Finds Shortest Path**: In all static environments with a clear path, the A* algorithm consistently identified the shortest path between the start and end nodes. This aligns with the theoretical guarantee of optimality when using an admissible heuristic (Manhattan distance in this implementation).
- **Impact of Obstacle Density**: As the density and complexity of obstacles increased, the number of nodes explored by the algorithm also increased, leading to a noticeable rise in computation time. This highlights the trade-off between environmental complexity and search efficiency.
- **Visualization of Search Process**: The Pygame visualization effectively demonstrated how A* intelligently explores the grid. Nodes closer to the optimal path were prioritized, and the `f_score` mechanism guided the search efficiently, avoiding unnecessary exploration of distant or blocked regions.

- **Path Representation:** As indicated in the provided notes, the path is ultimately represented as a sequence of coordinates, which is then visually rendered by coloring the nodes blue.

6.3 Performance Analysis

While precise quantitative metrics (e.g., exact computation time in milliseconds, number of nodes expanded) were not explicitly logged by the provided code, qualitative analysis and understanding of the algorithm's behavior allowed for a comparative performance assessment:

Feature/Algorithm	Basic A* (4-directional)	Weighted A* (Conceptual)	Dynamic A* (Conceptual)
Path Optimality	Optimal	Potentially Suboptimal	Optimal (with replanning)
Computational Speed	Moderate	Faster	Adaptive
Search Space	Explores fewer nodes than Dijkstra	Explores even fewer nodes	Reuses previous search data
Dynamic Environments	Not inherently suited	Not inherently suited	Well-suited
Heuristic Impact	Critical for efficiency	More pronounced	Integrated with replanning

- **Basic A*:** The implemented basic A* algorithm (with Manhattan distance) proved to be optimal and efficient for static environments. Its performance is superior to uninformed search algorithms and generally better than Dijkstra's in terms of nodes explored, especially in large grids.
- **Weighted A*:** Conceptually, if Weighted A* were implemented, it would significantly reduce the number of nodes explored by making the search more greedy. This would lead to faster path discovery but might result in a slightly longer path than the optimal one. This trade-off is often acceptable in applications where real-time responsiveness is paramount.
- **Dynamic A*:** For real-world applications with changing environments, the concept of Dynamic A* (or similar incremental search algorithms) is crucial. While not directly implemented, the ability to detect obstacles and recalculate paths locally would drastically improve adaptability and maintain near-optimal performance in dynamic

settings. This would involve more complex state management and efficient re-initialization of search parameters.

The experiments confirm that A* is a robust and versatile algorithm for path planning. Its effectiveness is further amplified by careful selection of heuristics and the incorporation of advanced techniques to handle more complex scenarios, such as dynamic obstacles and the need for faster, albeit slightly suboptimal, solutions.

7. Discussion

The comprehensive study and implementation of the A* pathfinding algorithm, along with its conceptual enhancements, highlight its significant role in autonomous navigation. The algorithm's strength lies in its heuristic-driven approach, which intelligently balances the cost incurred from the start node with an estimated cost to the goal, thereby optimizing the search process.

7.1 Real-World Applications

The versatility of A* makes it applicable across a broad spectrum of real-world domains:

- **Robotics Navigation:** A* is fundamental for robots to navigate complex environments, from industrial automation to domestic service robots. It enables robots to plan collision-free paths in warehouses, homes, and outdoor terrains.
- **GPS Routing Systems:** Modern GPS systems often employ variations of A* to calculate the fastest or shortest routes between locations, considering real-time traffic data and road networks.
- **Game AI Pathfinding:** In video games, A* is extensively used to enable non-player characters (NPCs) to navigate game worlds realistically, finding paths around obstacles and towards objectives.

7.2 Limitations and Considerations

Despite its advantages, the A* algorithm, particularly in its basic form, has certain limitations:

- **Static Environments:** The standard A* algorithm performs optimally in static environments where the map and obstacles are known and do not change. In dynamic environments, frequent replanning can be computationally intensive.
- **Heuristic Quality:** The efficiency of A* is highly dependent on the quality of the heuristic function. A poorly chosen heuristic can degrade performance, making it behave more like Dijkstra's algorithm (if the heuristic is too low) or Greedy Best-First Search (if the heuristic is too high and non-admissible).

- **Memory Consumption:** For very large grids or graphs, A* can consume significant memory to store the `open_set`, `closed_set`, and `came_from` dictionaries. This can be a constraint in resource-limited systems.

7.3 Impact of Heuristic Selection

The choice of heuristic function is a critical design decision that directly impacts both the efficiency and optimality of the A* algorithm. As demonstrated, Manhattan distance is appropriate for grid-based movement restricted to cardinal directions, guaranteeing optimality. However, for environments allowing diagonal movement, Euclidean distance becomes a more accurate and efficient heuristic. The concept of Weighted A* further illustrates how manipulating the heuristic's influence can trade off optimality for speed, a valuable consideration in time-sensitive applications. The balance between $g(n)$ and $h(n)$ is what makes A* so powerful, allowing it to adapt to different problem constraints and performance requirements.

This discussion underscores that while A* provides a robust foundation for path planning, its practical implementation and effectiveness are significantly enhanced by understanding its theoretical underpinnings and applying appropriate modifications for specific environmental and performance demands.

8. Conclusion and Future Work

This report has provided a comprehensive examination of the A* path planning algorithm, from its theoretical foundations to a practical Python implementation and discussions on advanced enhancements. The A* algorithm stands out as a powerful and versatile solution for pathfinding in various domains, owing to its intelligent combination of Dijkstra's optimality and Greedy Best-First Search's efficiency.

8.1 Key Conclusions

- **Optimality and Efficiency:** A* is proven to be optimal and efficient for pathfinding in static environments when coupled with an admissible heuristic. Its $f(n) = g(n) + h(n)$ evaluation function effectively guides the search towards the goal.
- **Heuristic Selection is Critical:** The choice of heuristic function (e.g., Manhattan vs. Euclidean distance) significantly impacts both the performance and the nature of the paths found. An appropriate heuristic is crucial for the algorithm's success.
- *Weighted A for Speed**: For applications where computational speed is prioritized over strict optimality, Weighted A* offers a viable solution by allowing a controlled trade-off between path length and search time.

- **Dynamic Obstacle Handling:** The ability to handle dynamic obstacles is paramount for real-world autonomous systems. While the core A* is designed for static environments, concepts like incremental replanning and local path recalculation are essential extensions.

8.2 Future Work

The insights gained from this study pave the way for several promising avenues of future research and development:

- **Integration with Machine Learning:** Future work could explore integrating A* with machine learning techniques. For instance, reinforcement learning could be used to learn optimal heuristic functions in complex, unknown environments, or to adapt path planning strategies in highly dynamic settings. This could lead to more robust and intelligent autonomous navigation systems.
- **3D Path Planning:** Extending the A* algorithm to three-dimensional environments is a critical step for applications involving aerial drones, underwater vehicles, or complex industrial spaces. This would involve adapting the grid representation, neighbor updating, and heuristic functions to account for the additional dimension.
- **Real-World Robot Implementation:** The current Pygame visualization provides a simulated environment. A crucial next step would be to deploy and test the A* algorithm on physical autonomous mobile robots. This would involve addressing challenges related to sensor integration, motor control, localization, and real-time performance under physical constraints and uncertainties.
- **Multi-Agent Pathfinding:** In scenarios with multiple robots or agents, coordinating their movements to avoid collisions and achieve collective goals efficiently presents a significant challenge. Future research could focus on developing multi-agent pathfinding algorithms based on A* that consider inter-agent dependencies and optimize global system performance.

This project serves as a strong foundation for continued exploration into advanced path planning strategies, contributing to the broader fields of robotics, artificial intelligence, and autonomous systems.

9. References

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10. Appendix: Source Code

Python

```
import pygame
import math
from queue import PriorityQueue

# Window size
WIDTH = 600
WIN = pygame.display.set_mode((WIDTH, WIDTH))
pygame.display.set_caption("A* Path Finding Visualizer")

# Colors
WHITE = (255, 255, 255)
BLACK = (0, 0, 0)
GREY = (128, 128, 128)
GREEN = (0, 255, 0)
RED = (255, 0, 0)
ORANGE = (255, 165, 0)
PURPLE = (128, 0, 128)
BLUE = (0, 255, 255)

# Node class
class Node:
    def __init__(self, row, col, width, total_rows):
        self.row = row
        self.col = col
```

```

        self.x = row * width
        self.y = col * width
        self.color = WHITE
        self.neighbors = []
        self.width = width
        self.total_rows = total_rows

def get_pos(self):
    return self.row, self.col

def is_barrier(self):
    return self.color == BLACK

def reset(self):
    self.color = WHITE

def make_start(self):
    self.color = ORANGE

def make_closed(self):
    self.color = RED

def make_open(self):
    self.color = GREEN

def make_barrier(self):
    self.color = BLACK

def make_end(self):
    self.color = PURPLE

def make_path(self):
    self.color = BLUE

def draw(self, win):
    pygame.draw.rect(win, self.color,
                      (self.x, self.y, self.width, self.width))

def update_neighbors(self, grid):
    self.neighbors = []
    # Down
    if self.row < self.total_rows - 1 and not grid[self.row + 1][self.col].is_barrier():
        self.neighbors.append(grid[self.row + 1][self.col])
    # Up
    if self.row > 0 and not grid[self.row - 1][self.col].is_barrier():
        self.neighbors.append(grid[self.row - 1][self.col])
    # Right
    if self.col < self.total_rows - 1 and not grid[self.row][self.col + 1].is_barrier():

```

```

        self.neighbors.append(grid[self.row][self.col + 1])
    # Left
    if self.col > 0 and not grid[self.row][self.col - 1].is_barrier():
        self.neighbors.append(grid[self.row][self.col - 1])

def __lt__(self, other):
    return False

# Heuristic (Manhattan Distance)
def h(p1, p2):
    x1, y1 = p1
    x2, y2 = p2
    return abs(x1 - x2) + abs(y1 - y2)

def reconstruct_path(came_from, current, draw):
    while current in came_from:
        current = came_from[current]
        current.make_path()
    draw()

def algorithm(draw, grid, start, end):
    count = 0
    open_set = PriorityQueue()
    open_set.put((0, count, start))
    came_from = {}

    g_score = {node: float("inf") for row in grid for node in row}
    g_score[start] = 0

    f_score = {node: float("inf") for row in grid for node in row}
    f_score[start] = h(start.get_pos(), end.get_pos())

    open_set_hash = {start}

    while not open_set.empty():
        for event in pygame.event.get():
            if event.type == pygame.QUIT:
                pygame.quit()

        current = open_set.get()[2]
        open_set_hash.remove(current)

        if current == end:
            reconstruct_path(came_from, end, draw)
            end.make_end()

```

```

        return True

    for neighbor in current.neighbors:
        temp_g_score = g_score[current] + 1

        if temp_g_score < g_score[neighbor]:
            came_from[neighbor] = current
            g_score[neighbor] = temp_g_score
            f_score[neighbor] = temp_g_score + h(neighbor.get_pos(), end.ge

            if neighbor not in open_set_hash:
                count += 1
                open_set.put((f_score[neighbor], count, neighbor))
                open_set_hash.add(neighbor)
                neighbor.make_open()

    draw()

    if current != start:
        current.make_closed()

    return False


def make_grid(rows, width):
    grid = []
    gap = width // rows
    for i in range(rows):
        grid.append([])
        for j in range(rows):
            node = Node(i, j, gap, rows)
            grid[i].append(node)
    return grid


def draw_grid(win, rows, width):
    gap = width // rows
    for i in range(rows):
        pygame.draw.line(win, GREY, (0, i * gap), (width, i * gap))
        for j in range(rows):
            pygame.draw.line(win, GREY, (j * gap, 0), (j * gap, width))


def draw(win, grid, rows, width):
    win.fill(WHITE)

    for row in grid:
        for node in row:

```



```

        node.draw(win)

    draw_grid(win, rows, width)
    pygame.display.update()

def get_clicked_pos(pos, rows, width):
    gap = width // rows
    y, x = pos
    row = y // gap
    col = x // gap
    return row, col

def main(win, width):
    ROWS = 30
    grid = make_grid(ROWS, width)

    start = None
    end = None

    run = True
    while run:
        draw(win, grid, ROWS, width)
        for event in pygame.event.get():
            if event.type == pygame.QUIT:
                run = False

            # LEFT CLICK
            if pygame.mouse.get_pressed()[0]:
                pos = pygame.mouse.get_pos()
                row, col = get_clicked_pos(pos, ROWS, width)
                node = grid[row][col]

                if not start and node != end:
                    start = node
                    start.make_start()

                elif not end and node != start:
                    end = node
                    end.make_end()

                elif node != end and node != start:
                    node.make_barrier()

            # RIGHT CLICK (reset)
            elif pygame.mouse.get_pressed()[2]:
                pos = pygame.mouse.get_pos()

```

```
row, col = get_clicked_pos(pos, ROWS, width)
node = grid[row][col]
node.reset()
```

```
if node == start:
    start = None
elif node == end:
    end = None
```

```
# PRESS SPACE → run algorithm
```

```
if event.type == pygame.KEYDOWN:
    if event.key == pygame.K_SPACE and start and end:
        for row in grid:
            for node in row:
                node.update_neighbors(grid)
```

```
        algorithm(lambda: draw(win, grid, ROWS, width), grid, start
```

```
# PRESS C → clear
```

```
if event.key == pygame.K_c:
    start = None
    end = None
    grid = make_grid(ROWS, width)
```

```
pygame.quit()
```

```
main(WIN, WIDTH)
```